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Chapter 25

GELFAND-P001186

16/IV–84 yr. Gelfand Sem.

Sinai. Ising model.

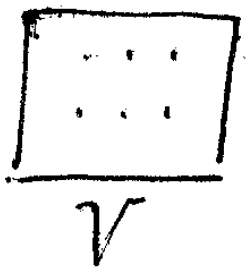
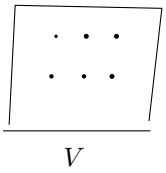


Diagram and typed description.



Original source details: GELFAND-P001186

$$\varphi(x) = \pm 1, \qquad H = -\sum_{\langle x,y\rangle} \varphi(x)\varphi(y),$$

$\langle x, y\rangle \leftarrow$ near. neighbors.

$$\frac{e^{-\beta H}}{\Xi(\beta \mid V)}, \qquad \Xi(\beta \mid V) = \sum_{\varphi(V)} e^{-\beta H}.$$

a) $\beta < \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\cdot\varphi(y)} \le Ce^{-\lambda\|x-y\|}, \quad \lambda = \lambda(\beta),$

b) $\beta > \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\varphi(y)} \longrightarrow m^2(\beta) > 0$ (long-range order),

c) $\beta = \beta_{\text{cr}} \Rightarrow \overline{\varphi(x)\varphi(y)} \sim \dfrac{C}{|x-y|^{\gamma}}, \quad \gamma = \dfrac{1}{4} \quad (d = 2).$



Uncertain reading. In line b), an additional stroke is visible inside the $>$ sign.

Original source details: GELFAND-P001186

In QFT a situation arises as with $\beta = \beta_{\text{cr}}$.

Explicitly computed for $d = 2$ are the free energy and the exponent γ.

— · —

Conjecture.

Let

$$\varphi_L(x) = \frac{\sum\limits_{y \in \Delta_L(x)} \varphi(y)}{L^\alpha}.$$

φ — rand. variables (see the Gibbs distr. above) with $\beta = \beta_{\text{cr}}$.

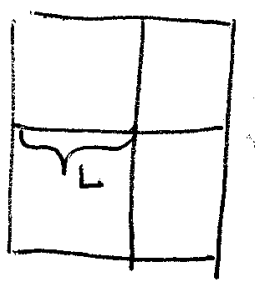
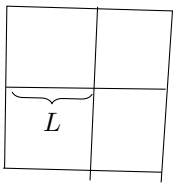


Diagram and typed description.



Original source details: GELFAND-P001186

Then the joint distribution of [inserted above: rand.] variables $\varphi_L(x)$ (at different x) has a limit and the lim. distr. does not depend on the model we started with.

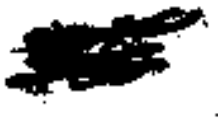
$$x = (x_1, x_2),$$

$$\Delta_L(x) = \{\, y = (y_1, y_2) \mid Lx_1 \le y_1 < L(x_1 + 1),\ Lx_2 \le y_2 < L(x_2 + 1) \,\}$$

contains L^2 points. For different x these squares do not intersect.

Renormalization group method — [crossed out: illegible?]

Uncertain reading. illegible



Original source details: GELFAND-P001186

eq. and fixed points for distr..

Scale transf. induce transf. [crossed out: on??]

Uncertain reading. on?



Original source details: GELFAND-P001186

in the set of probability distr..

For $d \geq 4$ [crossed out: illegible?]



Uncertain reading. illegible

Original source details: GELFAND-P001186

the method is trivial — there is a Gaussian fixed point.

Next d is treated as a cont. parameter [crossed out: illegible?].



Uncertain reading. illegible

Original source details: GELFAND-P001186

$\underset{4-\varepsilon}{4} \longrightarrow 3$ cont. $\Rightarrow$ some things are calc. for $d = 3$.

McPhirson Perverse[d?] shea[fs?]s etc.



Uncertain reading. English inscription: after Perverse, a retraced mark resembling d; after shea, several overlapping strokes precede the final s.

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(perverse sheaves)

GELFAND-P001187

23/IV Gelfand Sem. Serre. On the work of [crossed out: $G \ldots ?^?$]

Uncertain reading. $G \ldots ?$



Original source details: GELFAND-P001187

Delorme and Guichardet.

(1) Lemma. $A \subset$ aff. Hilbert H.

A bd. $\Rightarrow \exists!$ ball of min radius containing a.



Uncertain reading. The letter after "containing" resembles a lowercase a. The set is called A above.

Original source details: GELFAND-P001187

(min seq. ⇒ the centres converge)

(2) Theorem. [Mark crossed out: 1?[?]]



Uncertain reading. 1?

Original source details: GELFAND-P001187

A group of motions [inserted: in] aff. Hilbert sp. with [crossed out: illegible[?]]



Uncertain reading. illegible

Original source details: GELFAND-P001187

bd. orbits has a fixed point.

Proof: the centre of the min ball surrounding the orbit is fixed. ▶

Trees.

Prop. (FA): if G acts on a tree, then it has a fixed point.

[Note under (FA):]



Uncertain reading. arbre; a separate stroke sits above the beginning of the word.

Original source details: GELFAND-P001187

Examples 1) fin. [crossed out: illegible[?]]



Uncertain reading. illegible

Original source details: GELFAND-P001187

grp. 2) $SL_3(\mathbb{Z})$.

— · —

G. Margulis.

$$G = G_1 \times \ldots \times G_n \supset \Gamma \qquad \begin{matrix} G_i \text{ — simple} \\ \Gamma \text{ discrete} \end{matrix}$$

G/Γ is compact

or vol $G/\Gamma < \infty$.

Suppose there exist no such G', G'': $G = G' \times G''$

$$(G' \cap \Gamma) \text{ mult } (G'' \cap \Gamma) \quad \text{of fin. index.}$$

$$\text{rank } G \geq 2 \qquad (\text{rank } SL(n, \mathbb{R}) = n - 1)$$

$\Rightarrow \Gamma$ has prop. (FA).

— · —

GELFAND-P001188

3/IX–84 yr. Gelfand Sem.

I.M.G.

2 manif. M_1, M_2



Uncertain reading. A crossed-out or overwritten stroke is visible after M_2.

Original source details: GELFAND-P001188

are called Lipschitz equivalent, if $\exists$ a Lipsch. homeomorphism (rel. to 2 smooth metrics) (the inverse is also Lipsch.).

Example. surf. of a cube and a sphere.

Smoothness does not follow.

Sullivan's thm. On $\forall$ smooth manif. there is only one Lipsch. structure!!

(2 homeom. smooth manif. are Lipsch. equivalent)

Problem. Char. classes — integrals of curvatures.

g_{ik} must be C^2 fns.

(are 1st sufficient?)

How to express the classes in terms of g_{ik} without derivatives?

More precisely, it suffices to have the numbers, since only the res. of integration matters

(cf.: generalized fns.!)

Novikov's thm. on the agreement of char. numbers of homeom. manif. follows from Sullivan's thm..

— · —

$$\begin{pmatrix} \frac{\partial}{\partial x} & i\frac{\partial}{\partial y} \\ -i\frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{pmatrix}^2 = \begin{pmatrix} \Delta & 0 \\ 0 & \Delta \end{pmatrix}$$

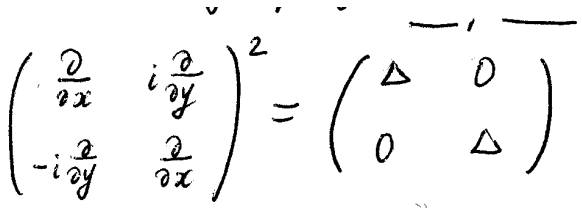
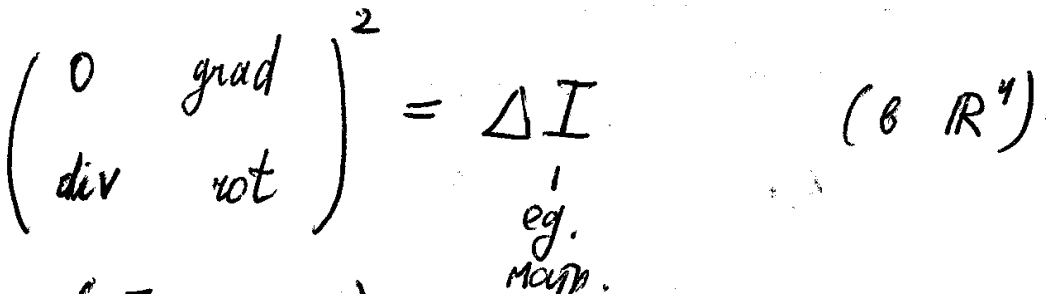


Diagram and typed description. Both diagonal entries of the matrix on the left are written without a minus sign.

Original source details: GELFAND-P001188

$$\begin{pmatrix} 0 & \text{grad} \\ \text{div} & \text{rot} \end{pmatrix}^2 = \Delta I \qquad (\text{in } \mathbb{R}^4)$$

[Under I: id. matr.]



Original source details: GELFAND-P001188

Diagram and typed description. The manuscript writes grad, div, rot and $\mathbb{R}^4$.

(Hamilton)

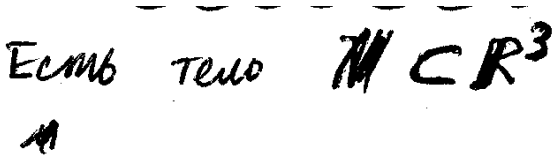
$$(d+\delta)^2 = \Delta \qquad (\delta = d^*)$$

(on forms)

GELFAND-P001189

The "almost everywhere" technique (Lebesgue).

There is a body [Overwritten: M or N?]?



Uncertain reading. [Overwritten: M or N?]

Original source details: GELFAND-P001189

$\subset R^3$. [Added below at left: A.] Volume — ? (via the areas of sections)

$S(x)$ — area of sections.

$$V = \int_a^b S(x)\,dx$$

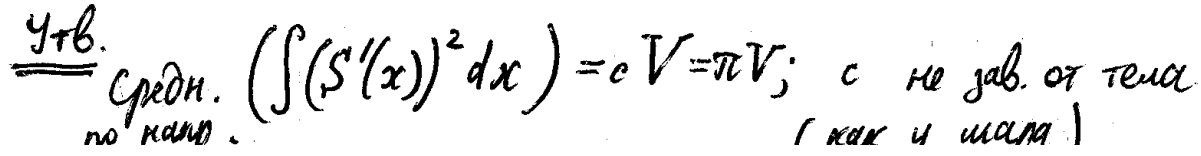
$\int (S'(x))^2\,dx$ also has the required dimension.

Average over all directions of the x-axis.

Assertion.

$$\underset{\text{over directions}}{\text{Mean}} \left(\int (S'(x))^2\,dx \right) = cV = \pi V;$$

c is independent of the body (as for a ball).



Original source details: GELFAND-P001189

Uncertain reading. The manuscript coefficient is π.

(see vol. 4 or 5 of Generalized Functions).

Problem. [Below the heading: (unsolved).] Take an arbitrary measurable set. Then $S(x)$ is measurable $(\forall x)$. Apparently, for almost all directions, $\exists S'(x)$. Question: in what sense is it differentiable, and what does the formula mean?

This is Sullivan's announcement.

On Grothendieck's preprint "Sketch of a Program".

Moved: Institute for Advanced Studies $\longrightarrow$ provincial university. Part has been written up as a 1,300-page manuscript.

Contents:

1. Message (dispatch)

2. The Lego–Teichmüller game and the Galois group Gal $\overline{Q}/Q$

[Below the name: Lego–Teichmüller.]

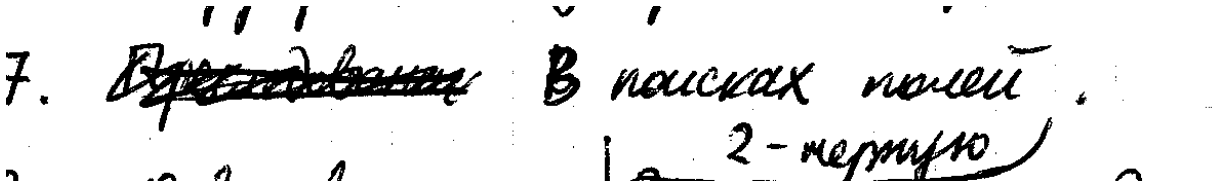
3. Number fields associated with a child's drawing.

4. ["Regular" crossed out and replaced above the line with a Russian synonym] polyhedra and finite fields.

5. Haro against general topology, and heuristic reflections on tame topology.

6. Differentiable and tame theory a là Nash.

7. [Crossed out: Pursuit?[?]]

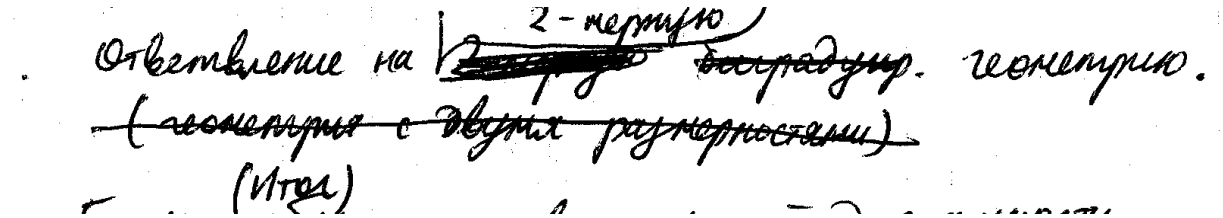


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Uncertain reading. Pursuit?

In search of fields.

8. Branching off into [Crossed out: two-dimensional? bigraded[?];



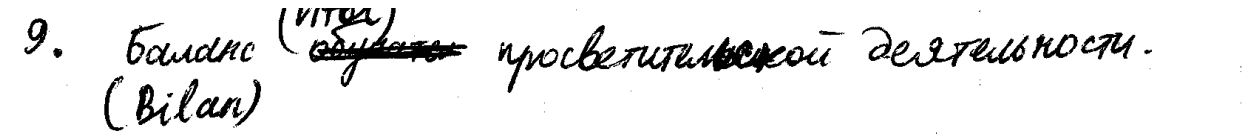
Original source details: GELFAND-P001189

Uncertain reading. two-dimensional? bigraded

inserted above: 2-dimensional] geometry.

[Line crossed out: (geometry with two dimensions).]

9. Balance sheet [crossed out: educational?[?];



Original source details: GELFAND-P001189

Uncertain reading. educational?

above the line: (Outcome)] of educational outreach.

[Below "Balance sheet": (Bilan).]

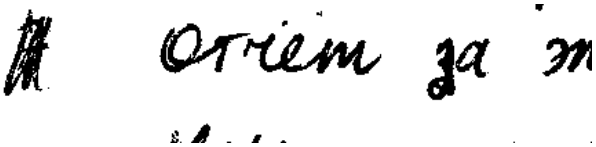
10. Epilogue.

GELFAND-P001190

[source number: 18]

In 1970 he stopped publishing.

[A sign crossed out in the left margin: И?[?]]



Uncertain reading. И?

Original source details: GELFAND-P001190

An account of the intervening period. Intuition of forms (geom.) precedes language.

(Topology and combinatorics — before differential)

Arnold.

Catalan numbers:

$$1, 1, 2, 5, 14, 42, \dots \qquad \text{(there are 34 definitions in all)}$$

$$\frac{(2n)!}{n!(n+1)!}$$

Harris (+ Mumford)

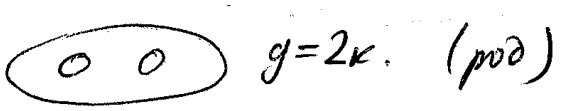


Diagram and typed description.

 $g = 2k$. (genus)

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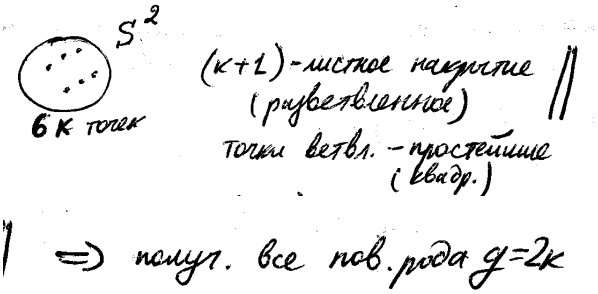
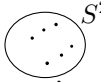


Diagram and typed description.

 S^2 $(k+1)$-sheeted covering (branched) //

$6k$ points

$\Rightarrow$ obtain all surfaces of genus $g = 2k$.

branch points — simple (quadratic).

Original source details: GELFAND-P001190

$3g$ parameters (in $\mathbb{C}$).

$SL(2)$ action $\Rightarrow 3g - 3$ moduli.

Number of times one obtains a general surface (without symmetries) — — Catalan numbers!!

— · —

$\Gamma \subset SU(2)$

[Below Γ, linked by a stroke: finite group.] [Vertical dividing stroke.] they come in types A_k, D_k, E_6, E_7, E_8.

McKay noticed (+ another connection of Arnold's)

(+ order of the Monster is related to the modular group)

Consider the symmetry group of a regular polyhedron, i.e. a finite subgroup $\Gamma \subset SU(2)$.

It has finite-dimensional representations.

We $\otimes$ with others and decompose. The resulting square matrix — — Cartan matrix (with an inessential correction) of the affine system.

Arnold:

cases; finite groups Γ act on $\mathbb{C}^2$; there is an algebra of invariants, and it has been described. There are 3 dependent invariants and one relation. [Below the line: (since the space is 3-dimensional).]

$$P(x, y, z) = 0$$

In $\mathbb{C}^3$ consider the fiber $\{P = \varepsilon\} = M_\varepsilon$, $\dim_{\mathbb{R}} M_\varepsilon = 4$. Consider the intersection matrix in 2-dimensional homology. This is the matrix of Coxeter. (FA, 1972, no. 4)

GELFAND-P001191

Groupoid: a category in which every morphism is an isomorphism.

Fundamental groupoid of a topological space:

points (objects), homotopy classes of paths (morphisms)



Diagram and typed description.



Original source details: GELFAND-P001191

Relative to a collection of points: when some initial points are disallowed.

10/IX–84 Gelfand seminar.

M. Semenov–Tian–Shansky.

On the conference (in Japan)! (near Kyoto)

H. / M.? Morimoto



Uncertain reading. H. / M.

Original source details: GELFAND-P001191

(8 young Japanese + 8 Western Europeans)

Yosida (young) — integrability

K. Taubs (California)

V. Kac

B. Kupershmidt

Wilson

[To the right of the list, beyond a vertical stroke:] P. [crossed out: Van?[?]]

Uncertain reading. Van?



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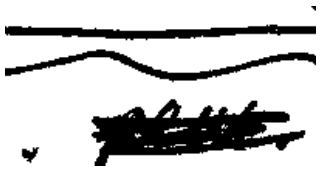
van Moerbeke. — criterion for integrability and nonintegrability based on Kovalevskaya's ideas.

Main outcome: open problems (to be published).

On P. van Moerbeke's talk (with ["someone" crossed out, "Adler" inserted above])

On integrability of [crossed out: solutions?[?]] algebraic differential equations.

Uncertain reading. solutions?



Original source details: GELFAND-P001191

In time — meromorphic functions.

They worked out the left-invariant metrics on $SO(4)$

Mishchenko, Fomenko + 2 more tops.

$$\mathbb{R}^n \subset \mathbb{C}^n \qquad \dot{z} = J_{(z)} \frac{\partial H}{\partial z} =: \{z, H\}$$

$J(z)$ — a polynomial defining a Poisson bracket satisfying the Jacobi identity.

The bracket may be degenerate.

Algebraically [inserted above the line: completely] integrable:

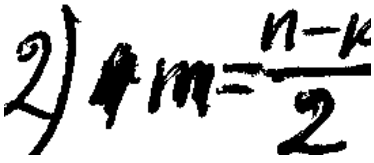
1) $\exists\, k$ polynomial invariants $H_1, \ldots, H_k$ (Casimirs)

[At right, arrow toward "Casimirs":] in the center of the Poisson bracket (they commute with everything) (polynomials in z)

[Lower right, another arrow toward "Casimirs":] center of the Lie algebra generated by the Poisson bracket.

2) [Overwritten sign: $\exists$?[?]] $m = \frac{n-k}{2}$ polynomial invariants with $\{H_i, H_j\} = 0$

Uncertain reading. $\exists$?



Original source details: GELFAND-P001191

Then $\forall c_i$ (almost all) [crossed out: $E_c \ldots ?^?$]

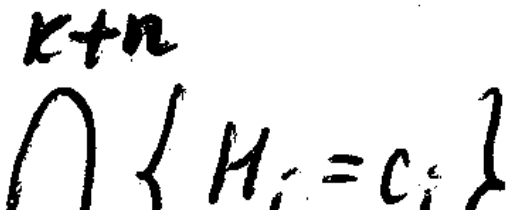
Uncertain reading. $E_c \ldots ?$



Original source details: GELFAND-P001191

$$\underbrace{\bigcap_{i=1}^{k+n} \{H_i = c_i\} \cap \mathbb{R}^n}_{E_c} \quad \text{— Liouville tori.}$$

Uncertain reading. The upper limit is written $k+n$.



Original source details: GELFAND-P001191

(an affine torus — either T^ℓ or $T^{\ell_1} \times \mathbb{R}^{\ell_2}$)

(Liouville–Arnold theorem)

Algebraically completely integrable.

Main theorem.

Algebraically completely integrable $\Rightarrow$ 1) solutions $z_i(t)$ are meromorphic in t; [crossed out: illegible$^?$]

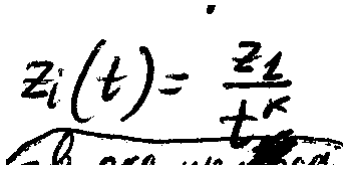
Uncertain reading. illegible



Original source details: GELFAND-P001191

$$z_i(t) = \frac{z_1}{t^k}$$

Uncertain reading. The numerator reads z_1, possibly z_i.



Original source details: GELFAND-P001191

[Circled insertion above “expansions”: Laurent.] [Circled below the formula, arrow left toward “expansions of solutions”: near the pole.]

expansions of solutions have [circled insertion: exactly] $(n-1)$ free parameters.

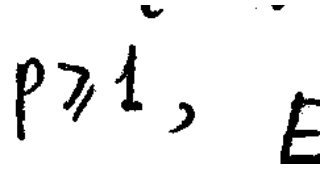
[At right, grouped with a brace:] $\exists\ (n-1)$-dimensional family of Laurent solutions — Laurent series with a pole at zero (or another point), satisfying the system.

2)

$$D_c = \left\{ z_i = \frac{z_i^0}{t^p} + \dots, \quad H_i(z(t)) = c_i + \dots \right\}$$

[Below $H_i(z(t))$: (have no pole at zero).]

$p \geq 1$,



Uncertain reading. Pole order letter: p or ρ.

Original source details: GELFAND-P001191

$E_c \cup D_c$ — an abelian set (canonical compactification).

D_i — an algebraic set in the space of $(n-1)$-parameter solutions, moreover $\longrightarrow$



Uncertain reading. The subscript on D in the last line looks like i; above it is c.

Original source details: GELFAND-P001191

GELFAND-P001192

[source number: 19]

Difficulty: [overwritten sign: illegible?]

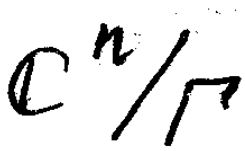


Uncertain reading. illegible

Original source details: GELFAND-P001192

constructing the formal series. Convergence is easier.

$$\longrightarrow \bigcap_i \{H_i = c_i\} \cup D_c = \mathbb{C}^n/\Gamma \qquad \text{(complex torus)}$$



Uncertain reading. The exponent is written n.

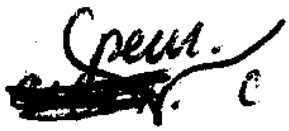
Original source details: GELFAND-P001192

Question: how is $\mathbb{C}^n/\Gamma$ related to the Jacobian (on which the equation is linearized).

They do not coincide!! But they differ by quotienting by a finite group.

The solution is written in terms of θ-functions.

In the practical construction the logic is reversed: we start with Laurent solutions depending on parameters. If ∃ [crossed out: illegible[?];



Uncertain reading. illegible

Original source details: GELFAND-P001192

above the line: Family] with $(n-1)$ free parameters, then ([crossed out: hypothesis?[?]] principle)



Uncertain reading. hypothesis?

Original source details: GELFAND-P001192

there is integrability. Moerbeke believes this is a general theorem.

— · —

[Crossed out: Example?[?]]



Uncertain reading. Example?

Original source details: GELFAND-P001192

I. M. G.

$$y'' = f(t, y, y')$$



Uncertain reading. The first argument is overwritten and reads as t.

Original source details: GELFAND-P001192

[An isolated sign to the right of the formula: a?[?]]



Uncertain reading. a?

Original source details: GELFAND-P001192

The singularities move. Painlevé equations: if the singularities are fixed.

Fixed singularities: for integrable systems. (Painlevé considered only essential singular points)

It is natural to regard as integrable those for which the essential singular points are fixed. Painlevé — order 2.

[Above the next line: Garnier — order 3 (?)]

Problem: give a definition of Painlevé-type integrability. There should be a classification.

S. T. S. Example.

In $\underbrace{\mathbb{R}^3}_{z_1,z_2,z_3} \oplus \underbrace{\mathbb{R}^3}_{z_4,z_5,z_6}$ there are 2 [crossed out: illegible?]



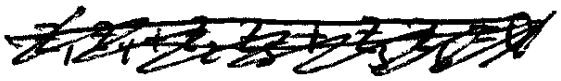
Uncertain reading. illegible

Original source details: GELFAND-P001192

sets of natural coordinates (Lie algebra $so(4)$)

[A semicolon separates the two coordinate annotations.] and z_{ij} (matrices)

[Line crossed out: $z_1z_4 + z_2z_5 + z_3z_6 = 0$?]



Uncertain reading. $z_1z_4 + z_2z_5 + z_3z_6 = 0$?

Original source details: GELFAND-P001192

$$H = \frac{1}{2}\sum \lambda_i z_i^2 + \sum_{i=1}^{3} \lambda_{i,i+3}\, z_i z_{i+3}$$

defines geodesics on $\underbrace{SO(4)}_{\text{6-dimensional}}$ (in a [inserted: left-] invariant metric); these are the equations for precession [inserted above: of momentum]; then one has to lift to $T^*SO(4)$.

[Two vertical strokes to the left of "6-dimensional."]

Only some are integrable. Some — Manakov; Mishchenko and Fomenko.

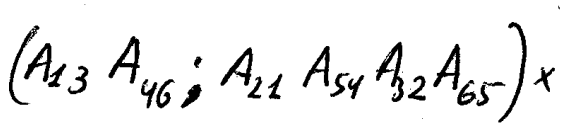
In each $\mathbb{R}^3$ the standard bracket (Lie–Poisson–Berezin–Kirillov)

$$\{z_i, z_j\} = \varepsilon_{ijk} z_k, \qquad i,k \in 1,2,3 \quad \text{or } 4,5,6$$

$$\{z_i, z_{j+3}\} = 0$$

2) $A_{ij} = \lambda_i - \lambda_j$

$$(\lambda_{14}^2, \lambda_{25}^2, \lambda_{36}^2)(A_{46}A_{32} - A_{65}A_{13}) = (A_{13}A_{46};\ A_{21}A_{54}A_{32}A_{65})\times$$



Uncertain reading. No separator is visible between A_{54} and A_{32}.

Original source details: GELFAND-P001192

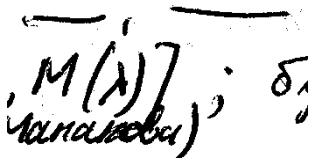
$$\times \left(\frac{(A_{25}-A_{36})^2}{A_{32}A_{65}}, \quad \frac{(A_{36}-A_{14})^2}{A_{13}A_{46}}, \quad \frac{(A_{14}-A_{25})^2}{A_{21}A_{54}} \right)$$

Here D — consists of 4 hyperelliptic curves (isomorphic) with the intersection described above.

Idea:

$$\dot{L}(\lambda) = [L(\lambda), M(\lambda)];$$

(Manakov's equation); we shall regard λ as an internal parameter.



Original source details: GELFAND-P001193

Uncertain reading. Above $M(\lambda)$ are two horizontal strokes and a short vertical mark.

$$\widetilde{sl}(n) = sl(n) \otimes \mathbb{C}(\lambda, \lambda^{-1})$$

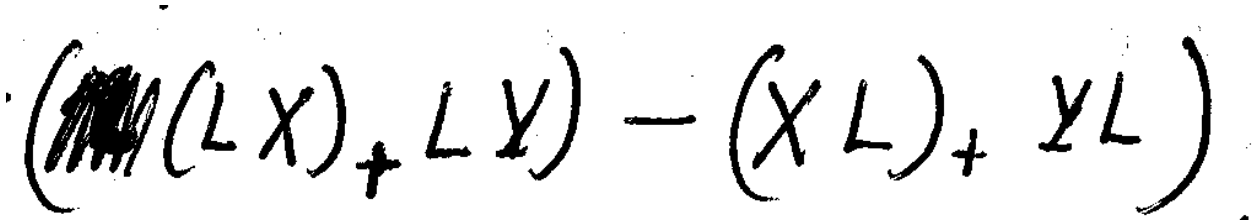
There are 2 brackets! The first is trivial (Berezin–Kirillov).

The second is constructed in R-matrix language

$$r = P_+ - P_-$$

$$\operatorname{Res}(\operatorname{tr} X \cdot Y)|_{\lambda=0}$$

$$\{f, g\}(L) = \big([\text{crossed out}]\,(LX)_+ LY\big) - (XL)_+ YL\big)$$

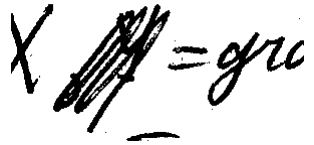


Original source details: GELFAND-P001193

Uncertain reading. The crossed-out fragment before (LX) is illegible. After the minus sign there is no visible opening parenthesis before $(XL)_+$.

(generalization of the Gelfand–Dickey bracket)

X [crossed out: Y?[?]]



Original source details: GELFAND-P001193

Uncertain reading. Y?

$=$ grad f, $Y =$ grad g.

For the series SO: the 2nd bracket cannot be obtained.

— · —

Drinfeld's theorem: description of the symplectic leaves in terms of double cosets in a larger Lie group.

$$G \times G \longrightarrow G \qquad \text{(Hamilton–Lie group)}$$

$d = g \oplus g^*$ — Lie algebra; g, g^* — subalgebras.

$$\{\text{orbits in } G\} = G^* \backslash D / G^*$$

Formula measuring the failure to preserve the Poisson bracket

$$X\{f,g\} - \{Xf, g\} - \{f, Xg\} = \langle X, [d_e \hat{f}, d_e \hat{g}]_* \rangle$$

This suffices to construct a variant of Hamiltonian reduction.

There is no general moment map.

On Grothendieck (Sosinsky). 3. Number fields associated with a child's drawing.

GELFAND-P001194

8/X–84. Gelfand seminar.

Beilinson on a paper by Atiyah and Singer

[At right, a curved line groups the title and journal line:]

"Dirac operators coupled to vector potentials."

Proc. Nat. Acad. Sci. USA,

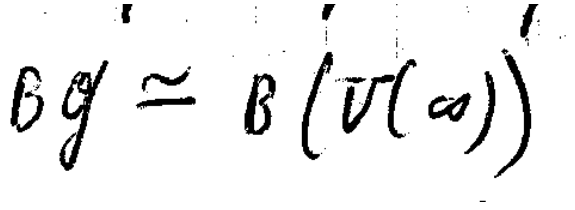
v. 81, pp. 2597–2600 (April 1984)

Construction of the classifying space of the gauge group.

Universal Dirac operator $\longrightarrow$ its index $\partial\!\!\!/ \in K_0(B\mathcal{G})$

$$B\mathcal{G} \simeq B(U(\infty))$$

$\Rightarrow$ on $B\mathcal{G}$ there is a virtual universal bundle — and this gives index $\partial\!\!\!/$.



Original source details: GELFAND-P001194

Uncertain reading. The letter in $B(U(\infty))$ reads as U, with a pointed lower bend.

$d \log(\det(\partial\!\!\!/^* \partial\!\!\!/))$ gives the class $t_1 \in H^1(\mathcal{G})$.

GELFAND-P001195

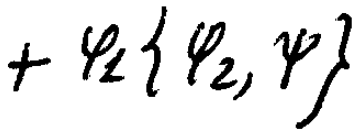
29/X–84 Gelfand seminar.

Semenov–Tian–Shansky M.

Drinfeld's theorem: a class of Poisson brackets.

M — set, suppose that on $C^\infty(M)$ there is a Poisson bracket $\{\varphi,\psi\}$ — a Lie algebra structure with the condition:

$$(1) \qquad \{\varphi_1\varphi_2,\psi\} = \varphi_2\{\varphi_1,\psi\} + \varphi_2\{\varphi_2,\psi\}$$



Uncertain reading. In the second term, the factor before the bracket is written φ_2.

Original source details: GELFAND-P001195

Then M is called Poisson.

This is a Category (morphisms — smooth maps preserving the brackets)

A Poisson Lie group G. — is a Lie group such that

1) G — a Poisson set

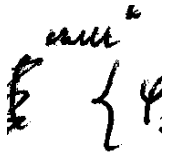
2)

$$\begin{aligned} G\times G &\longrightarrow G \qquad \text{— a Poisson map} \\ a,b &\longmapsto ab \end{aligned}$$

On $G\times G$ the direct product of brackets (there is a product in the category of Poisson sets)

$$\begin{aligned} \{\varphi(x,y),\psi(x,y)\} = {} & \{\varphi(\cdot,y),\psi(\cdot,y)\}(x) \\ & + \{\varphi(x,\cdot),\psi(x,\cdot)\}(y). \end{aligned}$$

[Above the line: or.] [At left, a crossed-out sign: {?$^?$]



Uncertain reading. {?

Original source details: GELFAND-P001195

$$\{\varphi_1,\varphi_2\}_{M_1\times M_2} = \{\varphi_1,\varphi_2\}_{M_1} + \{\varphi_1,\varphi_2\}_{M_2}.$$

Examples. 1°. G with the zero bracket

2°. g — a Lie algebra, g^* — its dual space; consider g^* as an additive Lie group. This is the Lie–Poisson group (with the canonical bracket!)

[Below "canonical bracket": Lie–Berezin–Kirillov.]

3°. G — a Lie group, where

(i) on the Lie algebra g there is an invariant inner product [Inserted above "invariant": nondegenerate.]

(ii) two subalgebras are given $g_+, g_- \subset g$, such that

$$g = g_+ \dot{+} g_-, \quad \text{and } g_+, g_- \text{ are isotropic}$$

(relative to the inner product).

GELFAND-P001196

Exercise. (I.M.G.) Suppose that on [inked over:



Undeciphered passage. The writing before E is obscured by heavy strokes.

Original source details: GELFAND-P001196

] E [below E, linked by a stroke: linear space] a Poisson bracket is introduced on linear functions. Then on E^* has a Lie algebra structure and the bracket on E is the one from the example.

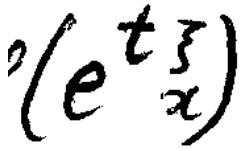
$$R \in \operatorname{End} \mathfrak{g}: \quad R = P_+ - P_-, \quad P_\pm \text{ — projections onto } \mathfrak{g}_\pm \text{ parallel to } \mathfrak{g}_\mp.$$

$$(2) \quad \{\varphi, \psi\} = (R(\nabla_\varphi), \nabla_\psi) - (R(\nabla'_\varphi), \nabla'_\psi)$$

[Callout under ∇_φ: left gradient $\in \mathfrak{g}$; under ∇'_φ: right gradient.]

Definition of the gradients.

$$(\nabla_\varphi(x), \xi) = \left(\frac{d}{dt}\right)_{t=0} \varphi(e_x^{t\xi})$$



Uncertain reading. x is written below $t\xi$, at the lower right of e. Possibly $e^{t\xi}x$.

Original source details: GELFAND-P001196

$$(\nabla'_\varphi(x), \xi) = \left(\frac{d}{dt}\right)_{t=0} \varphi(xe^{t\xi})$$

Theorem 1. (the proof is long). This bracket gives G the structure of a Poisson Lie group.

4°. $G = GL(n)$

$$\mathfrak{g} = \underset{\text{nilp.}}{\mathfrak{n}_+} + \underset{\text{diag.}}{\mathfrak{f}} + \underset{\text{nilp.}}{\mathfrak{n}_-}$$

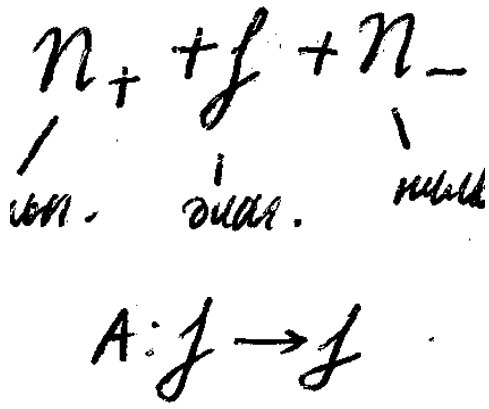
$$R = P_+ + AP_0 \text{ [inked over] } P_-,$$



Uncertain reading. The sign before P_- is heavily inked over. It could be $-$ or $+$.

Original source details: GELFAND-P001196

where $A\colon \mathfrak{f} \to \mathfrak{f}$ — a skew-symmetric operator.



Uncertain reading. The middle summand and both letters in the definition of A have the same shape. Rendered $\mathfrak{f}$, possibly $\mathfrak{h}$.

Original source details: GELFAND-P001196

$$((A, B) = \text{tr}(AB) \text{ on } gl(n))$$

Theorem 2 (in the setting of example 3°)
The symplectic leaves in G have the form

$$(G_+ x G_+) \cap (G_- y G_-) = M_{x,y};$$

(intersections of double cosets)
where $G_\pm$ — subgroups corresponding to the subalgebras $\mathfrak{g}_\pm$.
A Poisson set locally breaks up into leaves which are symplectic sets.
Symplectic leaves: locally a Poisson set looks like this

Diagram and typed description.
////



Original source details: GELFAND-P001196

(foliation; the leaves are symplectic, the bracket is recovered from the [inked over:



Undeciphered passage. The word before "symplectic" is heavily crossed out and remains illegible.

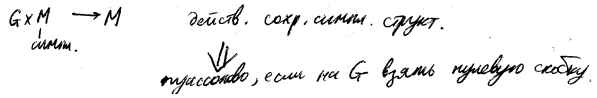
Original source details: GELFAND-P001196

] symplectic structure)

In fact there is a stratification! (the leaves can have different dim)

Def. Let G — a Poisson Lie group, M — a Poisson set; on $G \times M$ has a Poisson structure. We say that G acts on M in a Poisson way, if the action map $G \times M \to M$ is Poisson.

Such an action generally fails to preserve the Poisson bracket on M.



Original source details: GELFAND-P001196

Diagram and typed description.

$$\begin{array}{ccc} G \times M & \longrightarrow & M \\ | & & \\ \text{sympl.} & & \Downarrow \end{array}$$

the action preserves the symplectic structure

Poisson if we take the zero bracket on G

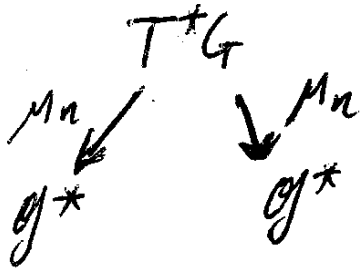
GELFAND-P001197

[source number: 34]

The condition that the action is Poisson is the equality

$$(3) \quad \{\varphi, \psi\}(g \cdot x) = \{\varphi(g\cdot), \psi(g\cdot)\}_M\big|_{\cdot=x} + \\ + \{\varphi(\cdot x), \psi(\cdot x)\}_G\big|_{\cdot=g}$$

The 2nd term measures the failure to preserve the Poisson bracket.



Original source details: GELFAND-P001197

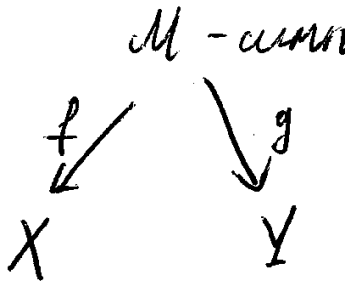
Diagram and typed description.

$$\begin{array}{ccccc} & & T^*G & & \\ & \swarrow^{\mu_\Pi} & & \searrow^{\mu_\Pi} & \\ \mathfrak{g}^* & & & & \mathfrak{g}^* \end{array}$$

Both arrow labels look like μ_Π. The text at right names the left and right maps.

— the left and right moment maps; they form a Poisson pair.

M — symplectic



Original source details: GELFAND-P001197

Diagram and typed description.

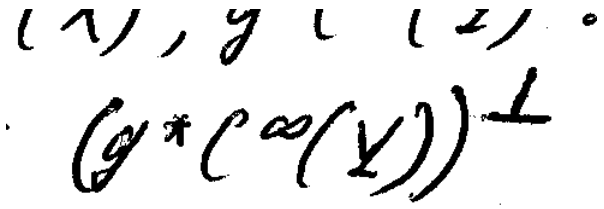
$$\begin{array}{ccccc} & & M & & \\ & \swarrow^{f} & & \searrow^{g} & \\ X & & & & Y \end{array}$$

— Poisson maps. They form a Poisson pair, if [in the right margin, with an arrow to "Poisson pair": or dual pair]

$$\{f^*C^\infty(X), g^*C^\infty(Y)\} = 0$$

Call the pair complete, if one of the algebras $f^*C^\infty(X)$, $g^*C^\infty(Y)$ is the centralizer of the other:

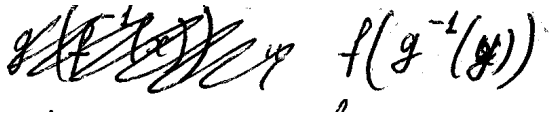
$$f^*C^\infty(X) = (g^*C^\infty(Y))^\perp$$



Uncertain reading. The symbol after the right parenthesis reads $\perp$.

Original source details: GELFAND-P001197

Theorem 3 (in the local form of the theorem). In this situation [crossed out: $g(f^{-1}(x))$] and $f(g^{-1}(y))$ — a symplectic leaf!



Uncertain reading. The letter inside g^{-1} is overwritten. A lower loop is visible and reads y, with a possible x beneath.

Original source details: GELFAND-P001197

(and locally all symplectic leaves are obtained this way)

We want to apply this to the proof of Drinfeld's theorem.

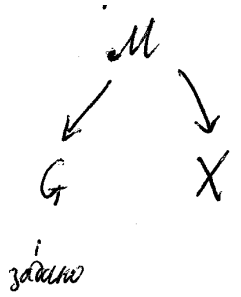
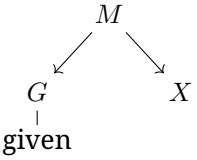


Diagram and typed description.



Original source details: GELFAND-P001197

For an arbitrary Poisson Lie group construct the double D (a group of twice the dim)

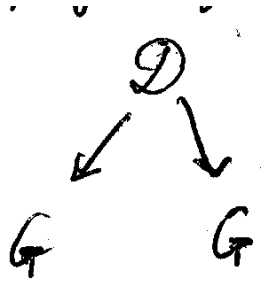
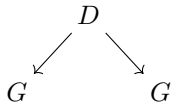


Diagram and typed description.



Original source details: GELFAND-P001197

$G = \mathfrak{g}^* \Rightarrow$ we obtain the Lie–Kirillov–Kostant construction

$$\mathfrak{d} = \mathfrak{g} \oplus \mathfrak{g}^*, \quad (\mathfrak{g}^* \text{ is also a Lie algebra, since there is a Poisson structure on } G)$$

D — the Lie group of the algebra $\mathfrak{d}$

On $\mathfrak{d}$ there is a natural inner product:

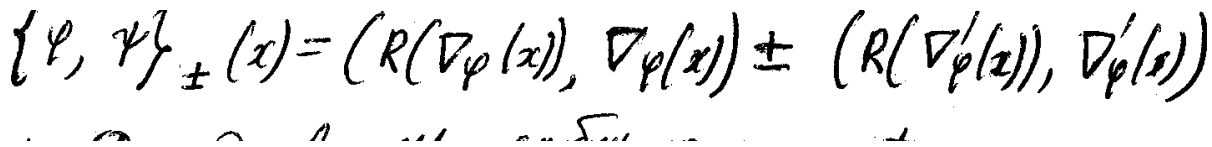
$$\langle (X, F), (X', F') \rangle = F(X') + F'(X)$$

This reduces to example 3°: the double acquires a Lie–Poisson group structure.

GELFAND-P001198

On G there are further 2 Poisson brackets:

$$\{\varphi,\psi\}_{\pm}(x) = (R(\nabla_\varphi(x)),\nabla_\varphi(x)) \pm (R(\nabla'_\varphi(x)),\nabla'_\varphi(x))$$



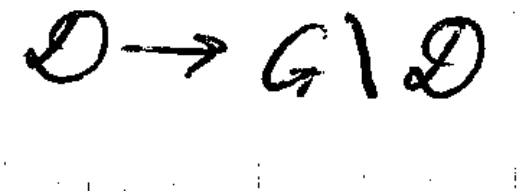
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Uncertain reading. All four subscripts on the right read φ. On the left, $\{\varphi,\psi\}$.

On D we should take the bracket with sign $+$.

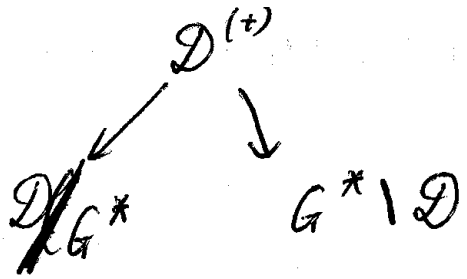
$\pm$ is related to homogeneity properties of the Sklyanin bracket.

Theorem 4. 1) Let G — be a Lie–Poisson group, D — its double. Then the canonical projections $D \to D/G^*$ and $D \to G\backslash D$ form a complete dual Poisson pair.



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Uncertain reading. The text has $G\backslash D$ with no asterisk. The diagram has $G^*\backslash D$.



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Diagram and typed description.

$$\begin{array}{ccc} & D^{(+)} & \\ \swarrow & & \searrow \\ D/G^* & & G^*\backslash D \end{array}$$

The slash at left is heavily overwritten and reads as /. The underlying strokes are indistinct.

2) $G \subset D/G^*$ and the structure is induced. [Under G, linked by a stroke: open cell.]

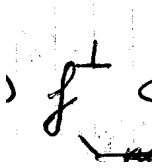
— · —

$H \subset G$ [Callout from H: subgroup.]

$$C^\infty(M)^H \subset C^\infty(M)$$

Thm. 5.

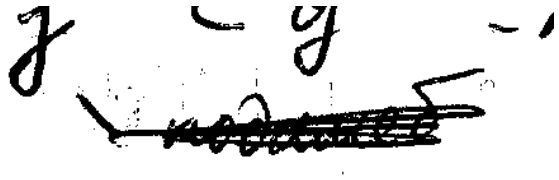
$$C^\infty(M)^H \text{ — Lie subalgebra} \iff \mathfrak{f}^\perp \subset \mathfrak{g}^* \text{ — Lie subalgebra}$$



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Uncertain reading. The base of $\perp$ has a lower loop and a cross stroke. Rendered $\mathfrak{f}^\perp$, possibly $\mathfrak{h}^\perp$.

[From the symbol with superscript $\perp$ a callout leads down to the crossed-out note]



Undeciphered passage. The note under the formula is heavily crossed out.

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Proof.

$$X\{\varphi,\psi\} - \{X\varphi,\psi\} - \{\varphi,X\psi\} = \langle [d_e\varphi(g\cdot x), d_e\psi(g\cdot x)]_*, X\rangle$$

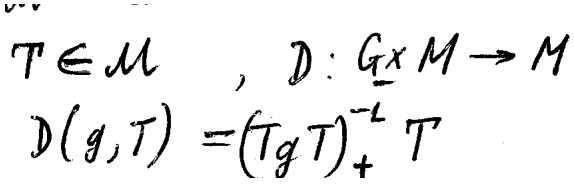
— · —

Let G — be the Poisson–Lie group from example 3°.

$M \subset G$ — a symplectic leaf

$$T \in M, \quad D\colon G_- \times M \to M$$

$$D(g,T) = (TgT)_+^{-1}T$$



Uncertain reading. The short stroke at the lower right of G reads as subscript $-$. Inside the parentheses is TgT. The exponent -1 appears only outside.

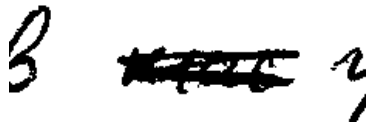
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(using the local factorization $x = x_+x_-$, $x_\pm \in G_\pm$)

Thm. 6. This action is Poisson

— · —

A quantum group is an object that in the classical limit goes over into [crossed out] group [abbr.].



Undeciphered passage. The word between "into" and "group" is obscured by cancellation.

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[Inserted below "goes over": algebra.]

(Drinfeld: group and cogroup. Both operations are series in h)

Sklyanin algebra: FA, 17:4